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Problem Set #2 (CAAM 37711)

(1) a - We define a neural architecture $g : \mathbb{R}^p \to \mathbb{R}^2$, where the input to g is a p-dimensional feature vector $x \in \mathbb{R}^p$. Thus, the output of the network is:

$$g(x) = \begin{pmatrix} g_1(x) \\ g_2(x) \end{pmatrix}, \text{ where } g_1, g_2 \in \mathbb{R} \text{ and } g \in \mathbb{R}^2.$$

- $f_\mu(x)$ is used to approximate $\mathbb{E}[Y|X] \in \mathbb{R}$. Since $f_\mu(x) \in \mathbb{R}$, we do not need to modify the data type of $f_\mu(x)$, and can just take the output link function to be the identity:

$$\boxed{f_\mu(x) = g_1(x)}$$

- $f_\sigma(x)$ is used to approximate $\sqrt{\mathbb{V}ar(Y|X)} \in \mathbb{R}^+$. Since the standard deviation must be positive but $g_2(x) \in \mathbb{R}$, we need to choose a link function that maps $\mathbb{R} \to \mathbb{R}^+$. This is naturally achieved through an exponential link function:

$$\boxed{f_\sigma(x) = e^{g_2(x)}}$$

b - We model data of the form in (a) as $Y|X \sim \mathcal{N}(f_\mu(x), f_\sigma(x)^2)$. The natural loss function for this model is the negative log-likelihood. For the Gaussian pdf

$$\mathbb{P}(y_i|x_i) = \frac{1}{\sqrt{2\pi} f_\sigma(x_i)} e^{-\frac{(y_i - f_\mu(x_i))^2}{2 f_\sigma(x_i)^2}},$$

the likelihood of the dataset (assuming the observations are independent) is given by

$$L = \prod_{i=1}^n \mathbb{P}(y_i|x_i) \longrightarrow \mathcal{L} = -\frac{1}{n}\log L = -\frac{1}{n}\sum_{i=1}^n \log\left[\mathbb{P}(y_i|x_i)\right],$$

where in the last step, we average on n samples to make the loss a per-sample average. Expanding the pdf above:

$$\mathcal{L} = -\frac{1}{n}\sum_{i=1}^n \left(\log\left[\frac{1}{\sqrt{2\pi} f_\sigma(x_i)}\right] + \log\left[e^{-\frac{(y_i - f_\mu(x_i))^2}{2 f_\sigma(x_i)^2}}\right] \right) = -\frac{1}{n}\sum_{i=1}^n \left[-\frac{1}{2}\log(2\pi) - \log(f_\sigma(x_i)) - \frac{(y_i - f_\mu(x_i))^2}{2 f_\sigma(x_i)^2} \right]$$

$$\longrightarrow \mathcal{L} = \frac{1}{n}\sum_{i=1}^n \left[\frac{1}{2}\log(2\pi) + \log(f_\sigma(x_i)) + \frac{(y_i - f_\mu(x_i))^2}{2 f_\sigma(x_i)^2} \right]$$

× we can drop the $\frac{1}{2}\log(2\pi)$ term, since it will not enter in the minimization procedure

Thus, the loss function we can use for this data is:

$$\mathcal{L} = \frac{1}{n}\sum_{i=1}^n \left[\log(f_\sigma(x_i)) + \frac{(y_i - f_\mu(x_i))^2}{2 f_\sigma(x_i)^2} \right] = \frac{1}{n}\sum_{i=1}^n \left[g_2(x_i) + \frac{1}{2}(y_i - g_1(x_i))^2 e^{-2 g_2(x_i)} \right]$$

penalizes inflated variances — noisy points count less

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(1) d - The model we have implemented suggests that, given the feature X of a trip:

$$\log Y \mid X = x \sim \mathcal{N}\left(f_\mu(x),\ f_\sigma(x)^2\right),$$

where $f_\mu(x)$ is the predicted mean and $f_\sigma(x)^2$ is the predicted variance. The statement, "what is the probability that a trip will last < 45 minutes" is translated into the language of our model as:

$$\mathbb{P}(Y < 45 \mid X = x) = \mathbb{P}(\log Y < \log 45 \mid X = x) = \Phi\left[\frac{\log 45 - f_\mu(x)}{f_\sigma(x)}\right],$$

where Φ is the CDF of the normal distribution.